

Using Machine Learning for Predicting Sleep Health & Lifestyle Patterns

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ABSTRACT

This paper presents a comparative evaluation of multiple machine learning models of Random Forest, XGBoost, Support Vector Machines (SVM), and Neural Networks for predicting sleep quality using lifestyle and physiological features. We investigate model behavior across both original and deduplicated versions of the Sleep Health and Lifestyle Dataset from Kaggle. Special attention is given to how evaluation metrics like average precision (AP) affect model performance in the presence of class imbalance. Our results show that models such as Random Forest and XGBoost achieve high accuracy and effective recall of minority classes, while SVMs and Neural Networks offer competitive results under the right conditions. Feature importance analysis highlights sleep duration, stress level, heart rate, and occupation as key predictors.

1 INTRODUCTION

Sleep quality is a cornerstone of health, influencing cognitive function, emotional regulation, and the risk of chronic illness. Modern lifestyles, stress, and physiological factors often disrupt sleep, making it crucial to understand and predict sleep quality for better health outcomes. This study uses machine learning techniques to model sleep quality based on lifestyle and biometric inputs, leveraging the Sleep Health and

Lifestyle Dataset. Our goal is to identify the most effective predictive models and understand which features most strongly influence sleep quality.

2 METHOD

2.1 Dataset Description

We used the Sleep Health and Lifestyle Dataset from Kaggle, which contains 374 synthetic records encompassing

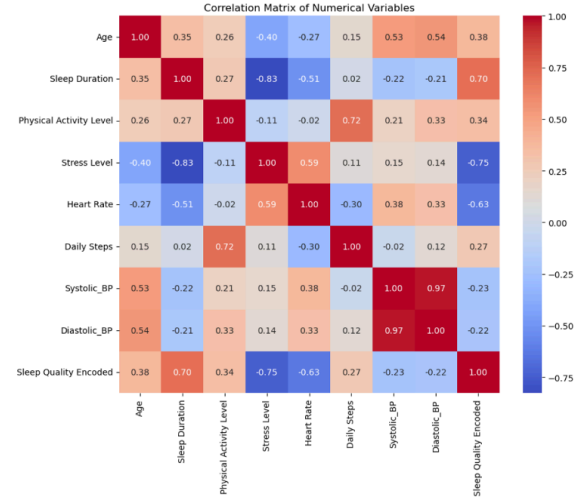
demographic, lifestyle, and biometric attributes such as Gender, Age, Occupation, Physical Activity Level, Stress Level, Sleep Duration, and Sleep Disorder. The target variable, originally scored on a 1 to 10 scale, was categorized into three sleep quality levels: Poor, Moderate, and Good.

Gender	Age	Occupation	Sleep Duration	Quality of Sleep	Physical Activity Level	Stress Level	BMI Category	Blood Pressure	Heart Rate	Daily Steps	Sleep Disorder	
0	Male	27	Software Engineer	6.1	6	42	6	Overweight	120/80	77	4200	None
1	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
2	Male	28	Doctor	6.2	6	60	8	Normal	125/80	75	10000	None
3	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea
4	Male	28	Sales Representative	5.9	4	30	8	Obese	140/90	85	3000	Sleep Apnea

To address concerns of data leakage and overfitting, we created a deduplicated version of the dataset, reducing the sample to 132 records. Both the original and deduplicated datasets were analyzed to assess the impact of duplicates on model behavior.

2.2 Preprocessing

Preprocessing involved imputing missing values (e.g., filling missing Sleep Disorder entries with 'None'), removing irrelevant identifiers, and decomposing the Blood Pressure field into Systolic and Diastolic values. Categorical variables were encoded using one-hot or ordinal methods, and numerical features were standardized. Correlation analysis and univariate plots highlighted key predictive features such as Sleep Duration, Stress Level, and Daily Steps.



SMOTE (Synthetic Minority Over-sampling Technique) was used to rebalance the training data. To ensure a consistent interpretation of feature importance across models, average precision was employed where appropriate.

A formula central to model evaluation was the Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

This was especially useful in tuning regression sub-components and understanding error magnitude.

Another important metric was Average Precision (AP), defined for multi-class problems as:

$$AP = \frac{1}{|C|} \sum_{c=1}^{|C|} Precision_c \times Recall_c$$

Where $|C|$ is the number of classes. This helped address the class imbalance issue inherent in our dataset. We also tuned model hyperparameters using GridSearchCV with cross-validation, which was key to achieving the reported results.

3 RESULTS & ANALYSIS

A Random Forest on the deduplicated dataset achieved 97% accuracy with high recall for minority classes. The most important features were Stress Level, Daily Steps, and Heart Rate. Switching the objective to average precision did not alter external results.

A Neural Network with ReLU activation and deeper architecture reached 90% accuracy, though it struggled with class 0 due to data scarcity. Re-tuning for AP improved internal validation but not test performance.

XGBoost also achieved 90% accuracy. Using AP as the evaluation metric raised it to 93% and identified daily steps, stress level, and occupation as key features.

SVM reached 90% accuracy and benefited from using a linear kernel. Under AP, performance improved to 93%. Occupation and stress level were the most important.

On the full dataset, all models performed better. Random Forest and Neural Network both reached up to 98% accuracy under AP tuning. XGBoost and SVM maintained 96% accuracy. In all models, duplicate records did not distort core feature importance.

4 CONCLUSION

Our comparative analysis shows that Random Forest and XGBoost consistently deliver strong accuracy and reliable minority class detection, particularly with SMOTE. Neural Networks and SVMs performed well given adequate tuning and data volume. Across all models, Daily Steps, Stress Level, Sleep Duration, and Occupation emerged as top predictors. AP optimization proved helpful for mitigating class imbalance and improving overall model robustness. These results affirm the feasibility of machine learning for sleep quality prediction.

REFERENCES

Colab Notebook Reference:

https://colab.research.google.com/drive/1F7Oqvgn6pYPArIhtxbbzQpmVsxB1_1h

Dataset Citation: Kaggle. (2023). Sleep Health and Lifestyle Dataset.

<https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset>